MATH 60646 – Stochastic Calculus
Term Project

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Question 2

What is the zero-coupon yield curve $y(t, T)$ and how is it related to the instantaneous risk-free rate r_t, the instantaneous forward rate $f(t, T)$ and the zero-coupon bond price $P(t, T)$? Describe clearly the economic meaning of each quantity and the relationship between them.

Answer 2(a)

1) Zero-coupon yield curve $y(t, T)$: annualized continuously-compounded return over $[t, T]$.

Intuition: the yield curve $y(t, T)$ represents the market-implied cost (interest rate) of borrowing/lending for different maturities T at time t.

2) Zero-coupon bond price $P(t, T)$: price at time t of 1 unit paid at T.

$$P(t, T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u \, du\right)\middle| \mathcal{F}_t\right].$$

Intuition: discount factor from T back to t.
Relations with y:

$$y(t, T) = -\frac{1}{T-t}\ln P(t, T), \qquad P(t, T) = \exp\big(-(T-t)y(t, T)\big).$$

Answer 2(b)

3) Instantaneous forward rate $f(t, T)$:

$$f(t, T) = -\frac{\partial}{\partial T} \ln P(t, T) \quad \Longleftrightarrow \quad P(t, T) = \exp\left(- \int_t^T f(t, u)\, du \right).$$

Intuition: market-implied infinitesimal rate for date T seen at time t.

Relation with y:

$$y(t, T) = \frac{1}{T - t} \int_t^T f(t, u)\, du$$

Answer 2(c)

4) Instantaneous risk-free rate (short rate) r_0:

$$r_0 = f(0,0) = -\frac{\partial}{\partial T} \ln P(0,0)\Big|_{T=0}$$

Intuition: current infinitesimal risk-free rate.

Relation with y (from/to short rate):

$$r_0 = \lim_{T \to 0} y(0, T)$$

Question 3

Let $(W_t)_{t\geq 0}$ be a standard Brownian motion under the physical measure $\mathbb{P}$. The short rate $(r_t)_{t\geq 0}$, representing the instantaneous risk-free rate, is assumed to follow the Cox-Ingersoll-Ross (CIR) square-root diffusion:

$$dr_t = \kappa^{\mathbb{P}} \left(\theta^{\mathbb{P}} - r_t\right) dt + \sigma \sqrt{r_t}\, dW_t^{\mathbb{P}}, \quad r_0 > 0.$$

(i) Why does this model admit a strong solution?

(ii) Do you have constraints on the parameters?

(iii) Can you interpret them?

(iv) What is the distribution of r_t?

(v) Describe the main mathematical properties of this short-rate dynamics. Explain why this specification is appropriate for modeling the evolution of the instantaneous rate.

Answer 3 (i): Strong Solution Existence I

$$dr_t = \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - r_t)dt + \sigma\sqrt{r_t}\, dW_t^{\mathbb{P}}.$$

Existence of strong solution: By *Itô's existence theorem*, a strong solution exists if the drift and diffusion coefficients satisfy:

1. **Lipschitz continuity of drift:** The drift $b(r_t) = \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - r_t)$ is linear in r_t and hence Lipschitz continuous.
2. **Linear growth of diffusion:** The diffusion coefficient $\sigma(r_t) = \sigma\sqrt{r_t}$ satisfies $|\sigma(r)| \leq C(1 + |r|)$, which is at most linear growth.
3. **Boundary condition at 0:** The *Feller condition* (to be discussed in part ii) ensures that the process does not cross zero and therefore the square root function remains well-defined along sample paths.

These conditions guarantee pathwise uniqueness and the existence of a unique strong $(\mathcal{F}_t)$-adapted solution.

Answer 3 (ii): Parameter Constraints

The key constraint is the **Feller condition**:

$$\boxed{2\kappa^{\mathbb{P}}\theta^{\mathbb{P}} > \sigma^2}$$

Additional (implicit) constraints:

- $\kappa^{\mathbb{P}} > 0$ (mean reversion is attractive, not repulsive)
- $\theta^{\mathbb{P}} > 0$ (long-run mean is positive)
- $\sigma > 0$ (volatility is positive)

Why Feller? When the diffusion coefficient $\sigma(r) = \sigma\sqrt{r}$ vanishes at $r = 0$, the natural scale analysis shows that zero is a *reflecting* boundary. The Feller condition ensures that the expected time a Brownian motion spends at zero is infinite, so in practice, if $r_t = 0$ occurs, the process instantly bounces back to positive values. Under the Feller condition, $r_t > 0$ with probability 1 for all $t > 0$.

Answer 3 (iii): Interpretation of Parameters

1. $\kappa^{\mathbb{P}}$ **(speed of mean reversion):** The parameter controlling how quickly deviations from $\theta^{\mathbb{P}}$ decay. Large $\kappa^{\mathbb{P}}$ means rapid mean reversion. Small $\kappa^{\mathbb{P}}$ means slow mean reversion (long-lived deviations).

2. $\theta^{\mathbb{P}}$ **(long-run mean or equilibrium level):** The unconditional steady-state level toward which the short rate is attracted. Over a sufficiently long horizon, $\mathbb{E}[r_t] \to \theta^{\mathbb{P}}$.

3. σ **(diffusion or volatility coefficient):** Controls the magnitude of random fluctuations. Typical units: $\sqrt{\text{basis points per year}}$. The square-root specification means volatility *decreases* when rates are low and *increases* when rates are high—a realistic feature.

Answer 3 (iv): Distribution of r_t

The CIR process has no closed-form explicit pdf, but r_t is distributed as a **scaled non-central chi-square random variable**.

Formal statement: For $t \geq s$,

$$r_t = b_t \chi^2_\nu(\lambda_c),$$

where

- $\chi^2_\nu(\lambda_c)$ is a non-central chi-square distribution with $\nu = \frac{4\kappa^{\mathbb{P}}\theta^{\mathbb{P}}}{\sigma^2}$ degrees of freedom
- $\lambda_c = \frac{4\kappa^{\mathbb{P}} r_s e^{-\kappa^{\mathbb{P}}(t-s)}}{\sigma^2(1-e^{-\kappa^{\mathbb{P}}(t-s)})}$ is the non-centrality parameter
- $b_t = \frac{\sigma^2(1-e^{-\kappa^{\mathbb{P}}(t-s)})}{4\kappa^{\mathbb{P}}}$ is the scaling factor

Moments under $\mathbb{P}$:

- $\mathbb{E}^{\mathbb{P}}[r_t \mid r_s] = r_s e^{-\kappa^{\mathbb{P}}(t-s)} + \theta^{\mathbb{P}}(1 - e^{-\kappa^{\mathbb{P}}(t-s)})$
- $\mathrm{Var}^{\mathbb{P}}[r_t \mid r_s] = \frac{\sigma^2}{2\kappa^{\mathbb{P}}} r_s (1 - e^{-\kappa^{\mathbb{P}}(t-s)}) e^{-\kappa^{\mathbb{P}}(t-s)} + \frac{\sigma^2\theta^{\mathbb{P}}}{2\kappa^{\mathbb{P}}}(1 - e^{-\kappa^{\mathbb{P}}(t-s)})^2$

Answer 3 (v): Mathematical Properties I

1. **Mean Reversion:** The deterministic drift $\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - r_t)$ pulls the process toward $\theta^{\mathbb{P}}$ at rate $\kappa^{\mathbb{P}}$. If $r_t > \theta^{\mathbb{P}}$, drift is negative (pulling down); if $r_t < \theta^{\mathbb{P}}$, drift is positive (pulling up). This ensures stationarity and prevents unbounded growth or decay.

2. **Positivity (Feller property):** Under the Feller condition, $r_t > 0$ almost surely for all $t > 0$, even if $r_0 = 0$. This is crucial because negative interest rates are economically undesirable (and historically were forbidden).

3. **Square-Root Volatility:** The diffusion term $\sigma\sqrt{r_t}\, dW_t^{\mathbb{P}}$ is proportional to $\sqrt{r_t}$. This means:
   - Volatility is high when r_t is high
   - Volatility is low when r_t is low
   - Volatility vanishes at $r_t = 0$ (reflecting barrier)

Answer 3 (v): Mathematical Properties II

Empirically, short-rate volatility is observed to be correlated with the level of rates, making this a highly realistic feature.

4. The CIR model admits closed-form bond prices and option prices (expressed in terms of noncentral chi-square distributions), making it practical for valuation and risk management.

Question 4 I

Under the risk-neutral measure $\mathbb{Q}$, the instantaneous risk-free rate is assumed to satisfy

$$dr_t = \kappa^{\mathbb{Q}} \left(\theta^{\mathbb{Q}} - r_t \right) dt + \sigma \sqrt{r_t}\, dW_t^{\mathbb{Q}}.$$

(i) Derive the link between the parameters $(\kappa^{\mathbb{P}}, \theta^{\mathbb{P}})$ and $(\kappa^{\mathbb{Q}}, \theta^{\mathbb{Q}})$.

(ii) Specify the functional form of the market price of risk that is required to ensure that the process remains of CIR type under $\mathbb{Q}$.

(iii) Prove that the Novikov condition is satisfied in this case.

(iv) Given that time series of the short rate r are not observable under the risk-neutral measure $\mathbb{Q}$, why is it common practice to maintain the same structural specification under $\mathbb{Q}$?

Question 5

Prove in a general framework that the discounted bond price process is a $\mathbb{Q}$-martingale.

Answer 5(a) – Setup And Martingale Definition

Fix maturity T and define the discounted bond price process

$$\widetilde{P}_t(T) := B_t^{-1} P(t, T), \qquad B_t := \exp\left(\int_0^t r_u \, du\right).$$

Under $\mathbb{Q}$, the bond pricing formula is

$$P(t, T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u \, du\right) \middle| \mathcal{F}_t\right].$$

Martingale definition ($(\mathcal{F}_t)$): a process (M_t) is a martingale if for all $0 \leq s \leq t$:

(i) M_t is $\mathcal{F}_t$-measurable,

(ii) $\mathbb{E}^{\mathbb{Q}}[|M_t|] < \infty$ (integrable),

(iii) $\mathbb{E}^{\mathbb{Q}}[M_t \mid \mathcal{F}_s] = M_s$ (martingale property).

Answer 5(b) – Verification Of All Properties

Using the pricing formula,

$$\begin{aligned}\widetilde{P}_t(T) &\overset{\text{def. } \widetilde{P}}{=} e^{-\int_0^t r_u\,du}\,\mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_u\,du}\middle|\mathcal{F}_t\right]\\ &\overset{\text{EC6}}{=} \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^t r_u\,du}e^{-\int_t^T r_u\,du}\middle|\mathcal{F}_t\right]\\ &= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_u\,du}\middle|\mathcal{F}_t\right].\end{aligned}$$

Let $X := e^{-\int_0^T r_u\,du}$ (it is nonnegative and integrable).

(i) $\widetilde{P}_t(T) = \mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_t]$ is $\mathcal{F}_t$-measurable (need proof?).

(ii) Integrable:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[|\widetilde{P}_t(T)|] &\overset{\text{def. of } \widetilde{P}_t(T)}{=} \mathbb{E}^{\mathbb{Q}}\left[\,\left|\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_t]\right|\,\right]\\ &\overset{\text{Jensen}}{\leq} \mathbb{E}^{\mathbb{Q}}\left[\,\mathbb{E}^{\mathbb{Q}}[|X| \mid \mathcal{F}_t]\,\right]\\ &\overset{\text{EC5}}{=} \mathbb{E}^{\mathbb{Q}}[|X|] < \infty.\end{aligned}$$

Answer 5(c) – Verification Of All Properties

(iii) Conditional expectation property: for $0 \leq s \leq t \leq T$,

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[\widetilde{P}_t(T) \mid \mathcal{F}_s] &\overset{\text{def. of } \widetilde{P}_t(T)}{=} \mathbb{E}^{\mathbb{Q}}\big[\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_t] \big| \mathcal{F}_s\big] \\ &\overset{\text{EC3}}{=} \mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_s] \\ &\overset{\text{def. of } \widetilde{P}_s(T)}{=} \widetilde{P}_s(T).\end{aligned}$$

Therefore $(\widetilde{P}_t(T))_{0 \leq t \leq T}$ is a $\mathbb{Q}$-martingale.

EC3: if $\mathcal{G}_1 \subseteq \mathcal{G}_2$ are σ-algebras, then $\mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1] = \mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_1]$.
EC5: $\mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}]] = \mathbb{E}^{\mathbb{P}}[X]$.
EC6: if Y is $\mathcal{G}$-measurable, then $\mathbb{E}^{\mathbb{P}}[XY \mid \mathcal{G}] = Y\,\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}]$.
Jensen used as: $|\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_t]| \leq \mathbb{E}^{\mathbb{Q}}[|X| \mid \mathcal{F}_t]$

Question 6 I

The time-t price of a zero-coupon bond maturing at T is

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s\,ds\right)\middle|\mathcal{F}_t\right].$$

Part (i): Prove that in the CIR model, the bond price admits the log-affine representation $\log P(t,T) = a(t,T) - b(t,T)\,r_t$ where

$$b(t,T) = \frac{2(e^{\gamma\tau}-1)}{(\gamma+\kappa^{\mathbb{Q}})(e^{\gamma\tau}-1)+2\gamma},$$

$$a(t,T) = \frac{2\kappa^{\mathbb{Q}}\theta^{\mathbb{Q}}}{\sigma^2}\log\left(\frac{2\gamma\,e^{(\kappa^{\mathbb{Q}}+\gamma)\tau/2}}{(\gamma+\kappa^{\mathbb{Q}})(e^{\gamma\tau}-1)+2\gamma}\right)$$

with $\tau = T - t$ and $\gamma = \sqrt{(\kappa^{\mathbb{Q}})^2 + 2\sigma^2}$.

Question 6 II

Part (ii): In the specific context of this model, derive the stochastic differential equation satisfied by the zero-coupon bond price $P(t, T)$.

Question 7 I

The limited number of risk-neutral parameters $(\kappa^{\mathbb{Q}}, \theta^{\mathbb{Q}}, \sigma)$ generally does not allow the model to match exactly the observed term structure of interest rates at time $t = 0$. As a consequence, the model may fail to reproduce the initial yield curve exactly. A common remedy consists in extending the model by replacing the constant long-run mean $\theta^{\mathbb{Q}}$ with a deterministic function of time, say $\theta_t^{\mathbb{Q}}$.

$$dr_t = \kappa^{\mathbb{Q}} \left(\theta_t^{\mathbb{Q}} - r_t \right) dt + \sigma \sqrt{r_t}\, dW_t^{\mathbb{Q}}, \quad r_0 > 0.$$

How can you choose $\theta_t^{\mathbb{Q}}$ so that you are fully consistent with the observed yield curve $y(0, T)$ at time $t = 0$?
You can retrieve the data for the zero-coupon yield curves from the Federal Reserve Economic Data (FRED) database: FRED. Provide a numerical implementation of the model and present the results

Question 7 II

with relevant graphs for a given day (each figure must include a detailed figure note providing all technical information necessary for proper interpretation of its content). How do you get parameter values for κ and σ?

Question 8

(i) Explain what is an interest rate swap (IRS).

(ii) Show that the time-t swap rate is a function of the zero-coupon bond prices at time t.

Question 9

(i) Explain what is a swaption.

(ii) Give the pricing formula of a swaption (with an explanation on how you obtain it).

References I

References will be added in the final version.